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AQA GCSE Geography (8035) — Paper 2: Challenges in the Human Environment · 3.2.1 Urban Issues and Challenges

2.1 Urban Issues in LICs and NEEs

1. Global Urbanisation Trends

Urbanisation: The process by which an **increasing proportion** of a country's population comes to live in urban areas (towns and cities). It is measured as the percentage of total population living in urban areas — NOT the total number of people in cities.

Key Global Statistics

Year	% Urban Population (global)	Urban Population (approx.)
1950	30%	750 million
2007	50%	3.3 billion (first time majority urban)
2018	55%	4.2 billion
2050 (projected)	68%	~6.7 billion

Urbanisation rates vary dramatically by development level:

Region	% Urban (2020)	Trend
North America	82%	Slow growth; some counter-urbanisation
Europe	75%	Slow growth; some re-urbanisation (gentrification)

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Latin America & Caribbean	81%	Largely already urbanised; growth slowing
Asia	51%	Rapid urbanisation ongoing — China, India, SE Asia
Africa	43%	Fastest urbanisation globally — doubling expected by 2050

Megacities

Megacity: A city with a population exceeding **10 million people**. In 1990 there were 10 megacities; by the mid-2020s there were over 30 (the UN counted 33 in 2025). Most growth in megacities now occurs in LICs and NEEs.

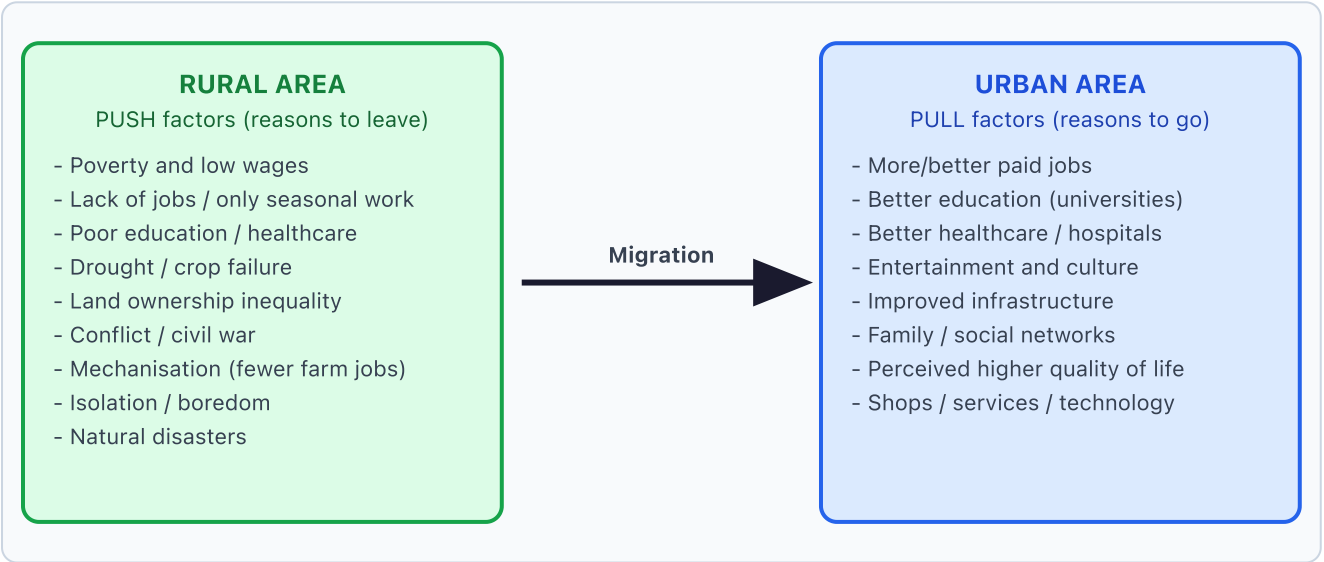
City	Country	Population (urban agglomeration)	Classification
Tokyo	Japan	~37.4 million	HIC — long ranked the world's largest city (a 2025 UN report using a new city definition ranked Jakarta and Dhaka larger)
Delhi	India	~32.9 million	NEE — one of the fastest growing megacities
Shanghai	China	~29 million	NEE
São Paulo	Brazil	~22.4 million	NEE
Mumbai	India	~21 million	NEE
Lagos	Nigeria	~15 million	NEE (lower-middle income) — one of Africa's largest cities
London	UK	~9.5 million	HIC

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Exam Tip: The distinction between urbanisation and urban growth is a classic exam trap. Urbanisation = increasing *proportion* living in cities. Urban growth = increasing *number*. A city can grow in absolute population without urbanisation occurring if the rural population grows at the same rate. State both clearly in high-mark answers.

2. Causes of Urbanisation

Urbanisation results from two processes: **rural-urban migration** (people moving from countryside to city) and **natural increase** (birth rate exceeding death rate within cities).



Push-Pull Factors in Depth — With Statistics

Push-pull theory (Ravenstein, 1885) explains migration as driven by negative conditions at origin and positive conditions at destination. In reality, migration decisions are complex and shaped by personal networks, distance, cost, and government policies.

Push Factor	Supporting Evidence
Agricultural mechanisation reducing farm jobs	In India, agricultural employment fell from 60% of the workforce in 2000 to ~42% by 2020 (World Bank), as millions left farming for urban jobs.

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Land inequality / landlessness	In Brazil in the 1970s–1990s, the top 1% of landowners held over 44% of agricultural land, pushing landless rural workers towards cities like São Paulo.
Climate shocks and drought	The 2009–2011 drought in the Sahel affected 10 million people; climate change could create 216 million internal climate migrants globally by 2050, 86 million of them in sub-Saharan Africa (World Bank, 2021).
Low rural wages	In Nigeria, rural farm incomes are typically far below urban earnings; the national minimum wage (₦30,000/month from 2019, raised to ₦70,000 in 2024) mainly applies to formal urban jobs, and even informal urban work often pays more than rural agriculture.
Conflict and instability	Syria's civil war (from 2011) displaced 6 million internally; many fled to Damascus, Aleppo, and eventually abroad. Northeast Nigeria's Boko Haram insurgency has displaced over 2 million people, mostly to towns and cities within the north-east such as Maiduguri.

Pull Factor	Supporting Evidence
Higher urban wages	The urban wage premium in developing countries is typically 15–30% above rural wages for equivalent work, and up to 200% for skilled urban jobs (ILO data).
Better education access	In Nigeria, urban areas have ~3x more secondary schools per capita than rural areas; university access is almost exclusively urban.
Healthcare	Under-5 mortality in rural Nigeria is much higher than in urban areas (NDHS, 2018) — a powerful pull for young families.
Social networks / family chain migration	Migrants follow established networks — a family member or friend who has already migrated reduces the cost, risk, and uncertainty of moving. This explains why migration streams often link specific rural regions to specific cities.
Perceived opportunity	Even the informal economy in Lagos or Mumbai offers more diverse income opportunities than subsistence farming — hawking, domestic work, construction, and petty trade all provide income that is irregular but potentially higher than rural work.

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Exam Tip: When asked about push-pull factors, avoid listing them without context. Examiners at higher grades reward analysis: explain *why* a factor is significant (e.g. "Mechanisation is particularly powerful because it permanently removes jobs, leaving no reason to stay — unlike drought, which is temporary"). Use the word "because" repeatedly to force explanation rather than description.

Natural Increase in Cities

In LICs and NEEs, cities also grow through **natural increase** — a higher birth rate than death rate within the urban population. Migrants are often young adults (reproductive age), which raises the urban birth rate. Improved healthcare in cities reduces child mortality compared to rural areas. In Lagos, for example, the natural increase rate is approximately 3% per year — meaning even without migration, the city would double in population within 23 years.

HIC vs LIC/NEE Urbanisation

Feature	HICs (e.g. UK, USA)	LICs/NEEs (e.g. India, Nigeria)
Stage of urbanisation	Already highly urbanised (75–85%); growth slowing or reversing	Rapidly urbanising; some cities growing at 3–5% per year
Main driver	Economic restructuring; suburbanisation/counter-urbanisation now prevalent	Rural-urban migration + high natural increase
Key challenge	Urban decay, deprivation, housing affordability, congestion	Insufficient housing, services, infrastructure for rapid population growth
Squatter settlements?	Rare / managed	Very common — ~1 billion people globally live in informal settlements
Historical timing	Urbanisation occurred gradually over 150+ years (Industrial Revolution from 1760s) allowing	Rapid urbanisation compressed into decades —

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	infrastructure to develop in parallel	infrastructure cannot keep up
Government capacity	Strong planning systems, building regulations, welfare provision	Often insufficient revenue, corruption, weak enforcement of planning law

3. Urbanisation in LICs/NEEs — Squatter Settlements

Squatter settlement / Informal settlement / Slum: A densely populated area of a city where housing has been **illegally built** on land not owned by the residents, using temporary materials, without official permission. Known by various local names: *favela* (Brazil), *barrio* (Latin America), *shanty town* (general), *bidonville* (French Africa), *bustee* (India).

Why Do Squatter Settlements Develop?

- Rural-urban migrants arrive in cities faster than formal housing can be built
- Migrants cannot afford formal sector rents or house prices
- Land on the city periphery, flood plains, steep hillsides, or near railways/factories is undesirable — squatters occupy it with less risk of eviction
- Governments in LICs lack resources to provide social housing at scale
- Informal settlements develop incrementally — start with temporary shelters, gradually improve over years

Characteristics of Squatter Settlements

Feature	Detail
Housing	Built from salvaged materials — corrugated iron, plastic sheets, cardboard, breeze blocks. Often 1–2 rooms for extended families.
Water supply	Often no piped water; residents buy from water vendors at 10× the price paid by formal city residents, or use shared/contaminated sources

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Sanitation	Few or no toilets/sewers; open sewers or defecation in the open; very high disease risk (cholera, typhoid, diarrhoea)
Electricity	Often illegally tapped from nearby power lines; fire risk; unreliable
Employment	Mostly informal economy — street vending, waste picking, domestic work, small manufacturing
Security	No legal title to land; constant threat of eviction; high crime rates in some areas
Health	High rates of respiratory disease (overcrowding, cooking fires); waterborne diseases; malnutrition; high infant mortality
Global scale	Approximately 1 billion people (1 in 7 globally) live in informal settlements (UN Habitat)

4. Case Study: Lagos, Nigeria (NEE) — Urbanisation in Africa

Case Study: Lagos, Nigeria — Africa's Megacity

Background and Location: Lagos is Nigeria's largest city, located on the Bight of Benin (Atlantic coast) in southwest Nigeria. It is the economic capital of Nigeria (though Abuja is the political capital) and the largest city in sub-Saharan Africa, with a metropolitan population of approximately 14–15 million (2023), growing at around 3% per year. By 2050, the UN projects Lagos will have over 30 million people.

Population Growth Timeline:

- 1950: ~290,000 people
- 1970: ~1.4 million (Lagos was Nigeria's capital until 1991 and its main port)
- 1990: ~5 million (rapid industrialisation; oil economy attracting workers)
- 2000: ~7.5 million
- 2015: ~12 million
- 2023: ~15 million (metropolitan area); ~24 million (urban agglomeration including surrounding areas)

Why Lagos grew so fast:

- Nigeria's oil boom from the 1970s concentrated wealth and government spending in Lagos
- Colonial legacy: Lagos was Nigeria's administrative and commercial centre under British rule
- Rural push from agricultural collapse: Nigeria went from a food exporter in 1960 to an importer by the 1980s; the oil boom destroyed agricultural competitiveness, pushing rural workers to cities
- Civil war (Biafra, 1967–1970) displaced millions; many eventually settled in Lagos
- High natural increase: Nigeria has a total fertility rate of ~5.3 children per woman (2020); urban birth rates remain high
- Informal economy opportunities: Lagos generates approximately 25–30% of Nigeria's total GDP despite having only ~8% of the population

Infrastructure Challenges:

- **Traffic:** Lagos is consistently ranked among the world's worst cities for traffic congestion (TomTom Traffic Index). Average commute times exceed 2 hours each way. The city's road network was designed for ~1 million people. The Lagos BRT (Bus Rapid Transit), opened 2008, carries 200,000+ passengers/day — Africa's largest BRT — but is overwhelmed.
- **Water:** Lagos Water Corporation officially supplies 210 million litres/day — but the city needs 700+ million litres/day. Most residents rely on private boreholes (often contaminated) or buy from water vendors at up to 10× the piped water price.
- **Electricity:** EKEDC (Lagos electricity distributor) provides power for an average of 8–12 hours/day in better-served areas, often far less. Most businesses run on diesel generators, adding to air pollution and costs.
- **Waste:** Lagos produces approximately 10,000 tonnes of solid waste per day; formal collection capacity is ~40% of this. The Olusosun landfill (one of the world's largest) has received waste for 30+ years and is now critically over capacity.
- **Flooding:** Much of Lagos is built on low-lying islands and coastal land. Annual flooding affects millions of residents; climate change is increasing flood frequency and severity.

Makoko — The Floating Slum: Makoko is one of Lagos's most well-known informal settlements, built on stilts over Lagos Lagoon. Key facts:

- Population estimated at 100,000–300,000 people (difficult to census accurately)
- Buildings are constructed on wooden stilts above the water; residents travel by canoe

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- No formal sanitation — sewage goes directly into the lagoon; the water is heavily polluted
- No piped water; residents use lagoon water (contaminated) or buy bottled/tanker water
- Children are rowed to school in canoes; there is a floating school (though the original structure collapsed in 2016 and a new one was rebuilt)
- Community is mostly from the Egun fishing community who have lived on the lagoon for generations — it is both an economic necessity and a cultural community
- In 2012, Lagos state government partially demolished Makoko with minimal notice, triggering international protests from human rights organisations. Many residents rebuilt.
- **Why it exists:** Land on Lagos Island is among the most expensive in Africa; waterfront land is formally controlled. The lagoon is unregulated space where poor migrants can build without land costs.

The Informal Economy in Lagos:

- An estimated 65–70% of Lagos workers are in the informal economy — street hawking, small trade, motorcycle taxis ("okadas"), domestic work, building and repair
- "Alaba International Market" — one of Africa's largest electronics markets — is largely informal
- The Lagos informal economy is estimated to be worth tens of billions of US dollars per year
- Okada (motorcycle taxis) provided critical transport for millions of poorer residents — the Lagos state government banned them from major roads in 2022, displacing hundreds of thousands of workers overnight, illustrating the fragility of informal livelihoods

Improvements and Government Response:

- Lagos BRT (2008) — Africa's largest Bus Rapid Transit system, 22 km route, 200,000+ passengers/day
- Eko Atlantic — a new city being built on reclaimed land adjacent to Victoria Island, designed for businesses and wealthy residents. Critics call it a "city for the rich" that does nothing for the urban poor.
- Lagos State has invested in CCTV, policing reform, and formal markets to reduce crime
- Drainage improvement projects to reduce flooding — though progress is slow relative to population growth

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Exam Tip — Lagos vs Mumbai: Lagos and Mumbai (both in NEEs) are often compared. Key contrasts: Nigeria's GDP per capita (around $1,000 or less in 2024, after the naira's devaluation) is now well below India's (~$2,700), and India's urban middle class is much larger. Lagos has less formal government capacity to fund improvement schemes than Mumbai. Lagos also faces the added challenge that Nigeria's federal government revenues are dominated by oil — which creates boom-bust cycles affecting infrastructure investment. In exam answers, specify whether a problem is LIC/NEE-specific or universal.

Lagos — Location, Importance and Opportunities

Importance: Lagos is Nigeria's main port (Apapa and Tin Can Island handle most of the country's imports), its commercial and financial centre, and home to the Nigerian Stock Exchange. Nollywood, Nigeria's film industry — one of the largest in the world by number of films — is based in and around Lagos. Murtala Muhammed Airport makes it a hub for West Africa.

Social opportunities:

- Better access to **services** than rural areas — more schools, universities (University of Lagos) and hospitals.
- Access to **resources** such as piped water and electricity, even if supply is unreliable, compared with rural areas that may have none.
- A diverse cultural and entertainment scene — music (Afrobeats), film and fashion industries.

Economic opportunities:

- **Jobs** in manufacturing, services, finance, oil-related industries and the port — far more varied than in rural areas.
- A booming **tech sector** — "Yabacon Valley" in Yaba hosts many start-ups and hubs.
- The informal economy provides work for millions who cannot find formal jobs, and gives a first foothold for new migrants.

Urban Planning Scheme for the Urban Poor — Makoko Floating School and Lagos slum upgrading

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- The **Makoko Floating School** (2013) was a prototype triangular building on floating plastic barrels, designed by architect Kunlé Adeyemi. It could rise and fall with the water and gave the community a school and meeting space that would not flood.
- It collapsed in a storm in 2016, but the design was refined into the MFS II and MFS III prototypes, and the idea of floating buildings has been promoted as a low-cost, flood-proof housing model.
- The **Lagos Metropolitan Development and Governance Project** (World Bank funded, 2006–2013) upgraded nine slum areas: drains were cleared to reduce flooding, roads paved, street lights installed and solid-waste collection improved for over a million people.
- **Evaluation:** such schemes improve quality of life cheaply and keep communities together, but they are small compared with the scale of need, and residents fear eviction because they lack legal land titles.

5. Case Study: Mumbai, India (NEE) — Squatter Settlements and Improvement

Case Study: Mumbai (NEE) — Urbanisation, Slums, and Improvement

Location and status: Mumbai (formerly Bombay) is India's financial capital, located on the west coast. It is India's largest city with ~12.5 million people in Greater Mumbai (2011 census) and over 20 million in the wider metropolitan region. It is one of the world's most densely populated cities. Around 40–50% of Mumbai's population lives in informal settlements occupying under 10% of the city's land area.

Why it grew so rapidly: Mumbai was the centre of British colonial trade (textiles, finance); post-independence it became India's financial and entertainment (Bollywood) hub. Millions migrated from rural Maharashtra, Uttar Pradesh, and Bihar seeking work and better wages.

Dharavi — Key Data: Located in central Mumbai, Dharavi covers approximately 2.1 km² and is home to an estimated 600,000–1 million people — making it one of Asia's most densely populated areas and one of its largest informal settlements.

- **Population density:** Up to 300,000 people per km² — approximately 10 times the density of Manhattan, New York

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- **Sanitation crisis:** 1 toilet per ~1,440 residents (compared to UN standard of 1 per 50)
- **Living space:** Most homes are 10–15 m² housing 5–10 people
- **Average income:** Approximately $500 per year — well below Mumbai's average of ~$3,000
- **Child health:** Infant mortality in Dharavi is approximately 68 per 1,000 live births vs. 36 per 1,000 for Mumbai as a whole
- **Education:** An estimated 25% of children in Dharavi do not attend school regularly; many work in family businesses

Dharavi's informal economy: Despite its challenges, Dharavi is a productive economic zone with an estimated annual turnover of **$600–700 million USD**. It contains ~15,000 single-room factories and 5,000 businesses including:

- **Recycling:** Processes up to 80% of Mumbai's plastic waste; workers (called "kabariwalas") sort and sell recyclables
- **Leather goods:** Many bags and accessories sold globally are produced in Dharavi workshops
- **Pottery (Kumbharwada):** Traditional pottery quarter has operated for over 100 years
- **Food production and garment manufacturing**

This demonstrates that informal settlements are not simply places of poverty — they are functional communities with complex economies.

Dharavi Redevelopment Project: A proposed redevelopment (approved in principle 2022, led by the Adani Group) aims to rehouse all residents in modern apartment blocks on the same site, while building additional flats for sale on the open market to fund the project. Critics argue: community businesses and social networks will be destroyed; many residents will not qualify for free housing due to technicalities; high-rise living may not suit residents' occupational needs (potters, leather workers need ground-floor space). Supporters say it will deliver piped water, sewage, electricity, and schools to all residents.

Squatter Settlement Improvement Schemes

Three main approaches are used by governments and NGOs to improve informal settlements:

Approach	How it works	Advantages	Disadvantages

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Self-Help Schemes (SHS)	Government provides materials (bricks, cement, roofing), legal land tenure, loans, and technical advice; residents build/improve homes themselves	Cheaper per family than government-built housing; community ownership and investment; preserves social networks; houses tailored to residents' needs	Quality depends on residents' skills; can be slow; corruption in allocation; may exclude the poorest who cannot contribute labour
Site and Service Schemes	Government provides serviced plots (road access, basic water supply, sewerage connection) and residents build their own home on the plot	More affordable than completed housing; residents invest in and maintain their own property; builds permanent community; used widely in East Africa and South Asia	Residents still bear building costs; initial site may lack full services; risk of subletting or resale to higher-income groups ("elite capture")
Slum Clearance / Forced Relocation	Government demolishes informal settlement and rehouses residents in purpose-built blocks elsewhere in the city	Can deliver high-quality permanent housing quickly; clears land for infrastructure development; improves area statistics	Destroys community and economic networks; new housing often far from residents' workplaces; high-rise flats unsuitable for families with outdoor businesses; residents may return to new slums; social trauma; often used to clear land for commercial development rather than residents' benefit

Mumbai — Slum Rehabilitation Authority (SRA) Data:

- Established 1995; has rehoused an estimated 600,000+ slum dwellers in permanent buildings under the Slum Rehabilitation Scheme
- The scheme provides free flats (raised to 300 sq ft / ~28 m² in 2016) to qualifying slum dwellers; developers are rewarded with additional "sale component" apartments for the market

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- Despite this, Mumbai still has thousands of slum pockets, home to an estimated 5–6 million people
- **Criticism:** Many residents are excluded for not meeting the "cutoff date" criteria (originally residents had to have lived there before 1 January 1995, later extended to 2000); the small flats are considered too small for families; buildings have had maintenance problems and lift failures

Exam Tip — Squatter Settlement Questions: Exam questions often ask you to "evaluate" improvement schemes. Always discuss at least two approaches and weigh pros vs. cons. The key evaluative point is that self-help and site-and-service schemes respect community agency, while slum clearance often serves developer or government interests more than residents. Support with specific Dharavi or Mumbai SRA data for top marks.

6. Urban Issues in LICs/NEEs

Issue	Detail	Example
Traffic congestion	Rapid motorisation as incomes rise; poor road infrastructure; mixed traffic (pedestrians, rickshaws, cars, buses); poor public transport	Mumbai: average commute speed 12 km/h in peak hours; Delhi: among the most congested cities globally (TomTom); Lagos: commuters spend average 4+ hours in traffic daily
Air pollution	Vehicle exhaust (older, poorly maintained vehicles); industrial emissions; domestic cooking fires; construction dust	Delhi: PM2.5 levels regularly 10–20× WHO safe limit; "AQI hazardous" for weeks in winter; air pollution was linked to about 1.7 million deaths in India in 2019. Lagos: vehicle emissions worsened by fuel quality (high-sulphur diesel)
Inadequate sanitation	Insufficient sewage infrastructure; open sewers; rivers used as open sewers	Dharavi: 1 toilet per 1,440 people; Mithi River in Mumbai severely polluted; Makoko (Lagos): sewage directly into lagoon
Water supply	Pipes inadequate; contamination; inequitable supply (rich areas get	Lagos: formal supply only 30% of need; Mumbai gets water for only 2–4 hours/day in many areas;

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	piped water, poor pay vendors); groundwater over-extraction	Jakarta sinking due to groundwater extraction
Waste management	Insufficient formal collection; illegal dumping; open landfills; burning of waste	Lagos: Olusosun landfill critically overcapacity; waste pickers (informal recyclers) handle much of the city's waste; Dharavi processes 80% of Mumbai's plastic waste informally
Crime and insecurity	Poverty, inequality, unemployment — especially youth unemployment — drives property crime and gang activity	São Paulo: gang-related violence; Lagos: "area boys" gangs controlling markets; rapid decline in murder rate in some favelas following targeted social programmes

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